Landmark Facility Solutions

A plain-English study guide — every module, explained, with the maths worked out

Financial Analysis of Corporate Deals • HBS 9-915-527

The two questions this case is about:

Broadway wants to buy Landmark for $120 million. (1) Is Landmark actually worth that to Broadway? (2) Should Broadway fund it with 100% debt, or a debt-and-equity mix? The twist: the value hinges on believing a turnaround, and the financing answer turns out to be the opposite of the usual reflex.

What's inside

Read it top to bottom the first time — each module builds on the one before.

- **Module 1 —** The big picture: two decisions, and what Landmark is really worth
- **Module 2 —** The discount rate: from comparable betas to Broadway's WACC
- **Module 3 —** Valuing Landmark to Broadway (three DCFs)
- **Module 4 —** Where the value comes from (the turnaround and the synergies)
- **Module 5 —** The financing decision: 100% debt vs a debt-and-equity mix
- **Module 6 —** Can Broadway actually service the debt?
- **Final results & exam watch-outs**

The story in one page

September 2014. Tim Harris runs Broadway Industries, a facility-services company on the East Coast (janitorial, HVAC, building maintenance). He wants to buy Landmark Facility Solutions, a larger facility-services company on the West Coast, for $120 million — which would turn Broadway into a coast-to-coast, fully integrated provider.

The problem with Landmark: it has great services and blue-chip clients, but its operating margin has collapsed from 3% in 2010 to under 1% in 2014. Harris is convinced this is bad management, not a broken business — replace the team, cut the bloat, and the margin recovers. The whole valuation rests on whether you believe him. Two board members don't, and his own advisors are nervous.

The numbers you'll keep using:

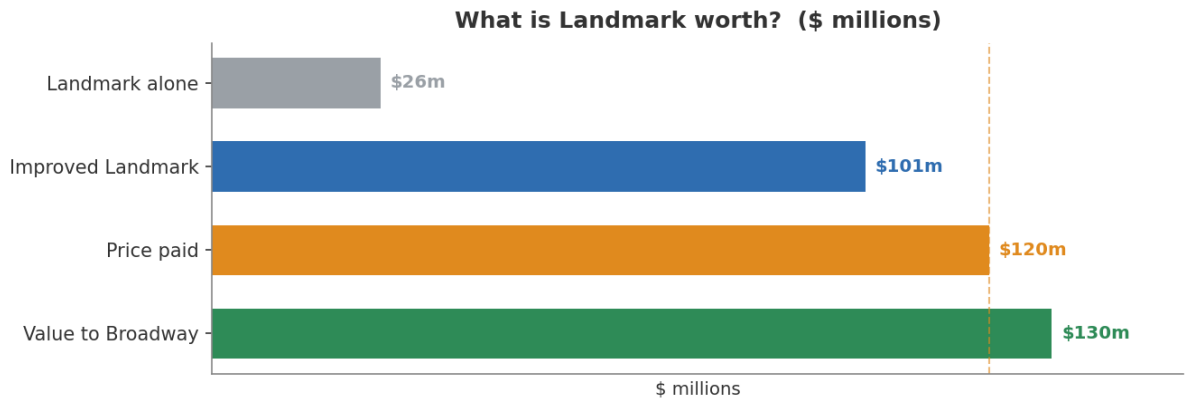
Item	Value
Asking price for Landmark	$120 million
Corporate tax rate (both firms)	35%
Market risk premium	5.9%
Risk-free rate (10-yr Treasury)	2.56%
Comparable companies' equity betas	1.69 / 1.25 / 1.56
Terminal growth after 2019	3–4%
Broadway's existing debt	$7.9 million

Module 1 — The big picture

Two separate decisions are tangled together here, and it helps to keep them apart:

- **Should Broadway buy Landmark at all?** i.e. is Landmark worth at least $120m to Broadway? (Modules 2–4.)
- **If yes, how should it pay — 100% debt or a debt-and-equity mix?** (Modules 5–6.)

Here is the answer to the first decision in one picture. The key is that Landmark is worth very little on its own, but a lot to Broadway — because only Broadway can fix the margins and layer on synergies.



Landmark on its own is worth only ~$26m. But fixed up and combined with Broadway it's worth ~$130m — just above the $120m price. The gap between $26m and $130m is the whole debate: it only exists if the turnaround is real.

The tension that runs through the case:

Landmark's sub-1% margin is either bad management (fixable, so the company is a bargain) or a broken business model (not fixable, so $120m is far too much). Harris bets on the first. Every number below assumes the turnaround works — which is exactly what to stress-test.

Module 2 — The discount rate

Before valuing anything, we need the rate to discount future cash flows. We can't use the comparable companies' betas directly, because each carries a different amount of debt, and debt inflates beta. So the standard three-step move: strip the debt out of each comparable's beta (unlever), average them, then re-load Broadway's target debt (relever).

Step 1 — unlever each comparable's beta

Unlevered beta = levered beta × Equity ÷ [Equity + (1 − tax) × Debt]. With a 35% tax rate:

Comparable	Levered β	Equity ($m)	Debt ($m)	Unlevered β
Company 1	1.69	6,186.9	5,887.0	1.04
Company 2	1.25	3,151.7	355.0	1.16
Company 3	1.56	500.1	289.0	1.13
Average	—	—	—	**1.11**

Step 2 — the unlevered cost of capital (CAPM)

Using the average asset beta of 1.11, the risk-free rate of 2.56% and the 5.9% market risk premium:

Unlevered cost of capital = 2.56% + 1.11 × 5.9% = 9.13%

Step 3 — relever and get the WACC

Re-load the industry's debt (average Debt/Equity ≈ 0.55), which lifts beta back up, then run CAPM again for the cost of equity, and blend with after-tax debt:

Re-levered beta = 1.11 × [1 + (1 − 0.35) × 0.55] = 1.51
Cost of equity = 2.56% + 1.51 × 5.9% = 11.47%
WACC = 0.65 × 11.47% + 0.35 × (1 − 0.35) × 5.5% = 8.68%

So every DCF in this guide discounts at **WACC = 8.68%**. (The unlevered 9.13% is the rate you'd use for a pure-play, debt-free version; Prof. Dash values the businesses at the levered WACC of 8.68%.)

Module 3 — Valuing Landmark to Broadway

The value of Landmark to Broadway comes in two parts, and there's a base-case comparison, so it takes three DCFs. All three use the same recipe — free cash flow = NOPAT + depreciation − change in working capital − capex — discounted at 8.68%, with a terminal value at the end.

DCF 1 — the improved Landmark (the turnaround)

Under Broadway, Landmark's margin climbs from 1.5% toward 3%, and working capital is squeezed from 7% down to 5.5% of sales. In $ millions:

$ million	2015	2016	2017	2018	2019
Revenue	362.8	380.9	400.0	420.0	441.0
Operating profit (margin → 3%)	5.4	7.6	10.0	12.6	13.2
NOPAT (after 35% tax)	3.5	5.0	6.5	8.2	8.6
+ Depreciation	2.1	2.4	2.7	3.0	3.3
− Change in working capital	(0.1)	(0.6)	0.4	0.0	(0.9)
− Capital expenditure	3.6	3.8	4.0	4.2	4.4
= Free cash flow	**2.1**	**4.2**	**4.8**	**7.0**	**8.4**

Terminal value (at end of 2019) = next year's FCF ÷ (WACC − growth) = 7.0 ÷ (8.68% − 3%) ≈ $123m. Discount the five cash flows and the terminal value at 8.68% and you get the value of the improved Landmark: **$100.9 million.**

Notice: that's below the $120m price. The improved Landmark alone doesn't justify the price — you need the synergies too.

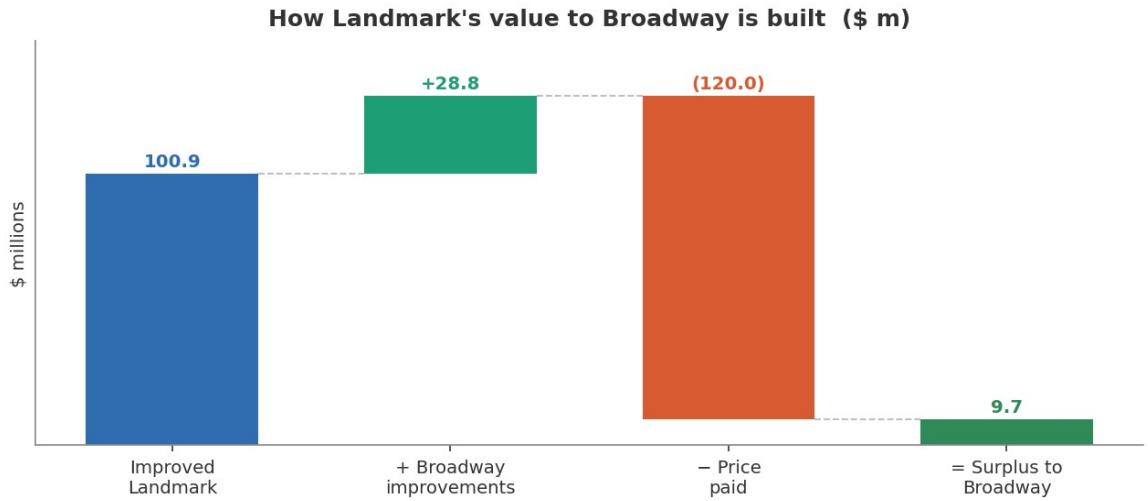
DCF 2 and 3 — Broadway with vs without the deal

The acquisition also lets Broadway apply a premium-pricing strategy to its own business. So we value Broadway two ways and take the difference:

- **Broadway with the acquisition** (premium pricing, higher margins): worth $86.2m.
- **Broadway without the acquisition** (business as usual): worth $57.4m.
- **The improvement to Broadway's own business** = 86.2 − 57.4 = $28.8m.

Adding it up

Piece	Value
Landmark on its own (no turnaround)	$26.0m
Improved Landmark (run by Broadway)	$100.9m
+ Improvements to Broadway's own business	$28.8m
= Total value of Landmark to Broadway	**$129.7m**
− Price Broadway pays	$120.0m
= Surplus (value created, before financing)	**$9.7m**



The improved Landmark ($100.9m) plus the boost to Broadway's own business ($28.8m) is worth $129.7m — $9.7m more than the $120m price. That $9.7m surplus only exists if the turnaround and the premium-pricing strategy both work.

Module 4 — Where the value comes from

The jump from Landmark's $26m standalone value to $129.7m to Broadway is not magic — it's four concrete sources. Being able to name them (and doubt them) is what an exam answer needs.

- **Fixing the margins.** Cutting excess executive pay, perks, and non-essential marketing lifts Landmark's operating margin from under 1% toward 3%. This is the single biggest driver — and the most debatable.
- **Eliminating duplicate overhead.** One head office instead of two: shared executives, support staff, and office space.
- **Releasing working capital.** Applying Broadway's tighter processes drops Landmark's non-cash working capital from ~7.4% of sales toward 5.5% — a one-off cash release.
- **Premium pricing.** Using Landmark's respected brand to charge more across the combined firm. This is the boldest assumption: it actually shrinks Broadway's own revenue by ~10% a year for two years before margins recover — which is exactly why the board is skeptical, and a fair thing to challenge.

The honest caveat (worth a sentence in any answer):

Every source of value depends on management execution and on customers accepting higher prices. If the turnaround only half-works, the $129.7m shrinks fast and the $9.7m surplus can vanish. The valuation is only as good as the belief in the plan.

Module 5 — The financing decision

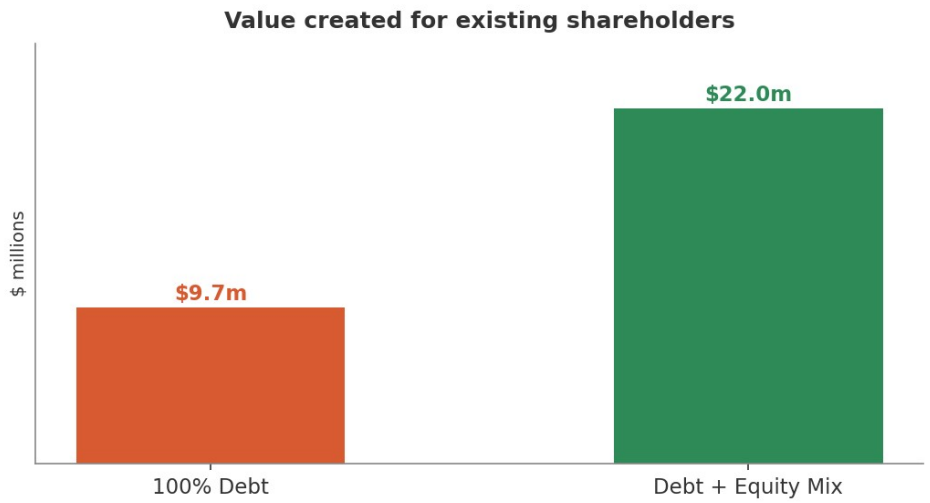
Broadway can fund the $120m purchase two ways. This is the heart of the case, and the answer is not the obvious one.

Option 1 — 100% Debt: $120m loan at 5.5%, 9-year; principal amortized $5m/year from 2017 with a $90m balloon at maturity (2023).

Option 2 — Debt + Equity Mix: $60m loan at 5% (repaid in full at maturity, 2020) plus $60m of new equity for 40% of the combined firm. Harris's stake falls from 100% to 60%. New equity holders get no dividends until the debt is repaid.

The combined firm is worth the same either way — about $187.1m. So the only question is how that value gets split between lenders, existing owners, and new investors. Equity value = combined value − debt; then split the equity.

	100% Debt	Debt + Equity Mix
Combined firm value	$187.1m	$187.1m
− Total debt (new + existing $7.9m)	$127.9m	$67.9m
= Equity value	**$59.2m**	**$119.2m**
Existing shareholders keep	100% → $59.2m	60% → $71.5m
New investors get	—	40% → $47.7m
Existing equity before the deal	$49.5m	$49.5m
Value created for existing shareholders	**$9.7m**	**$22.0m**



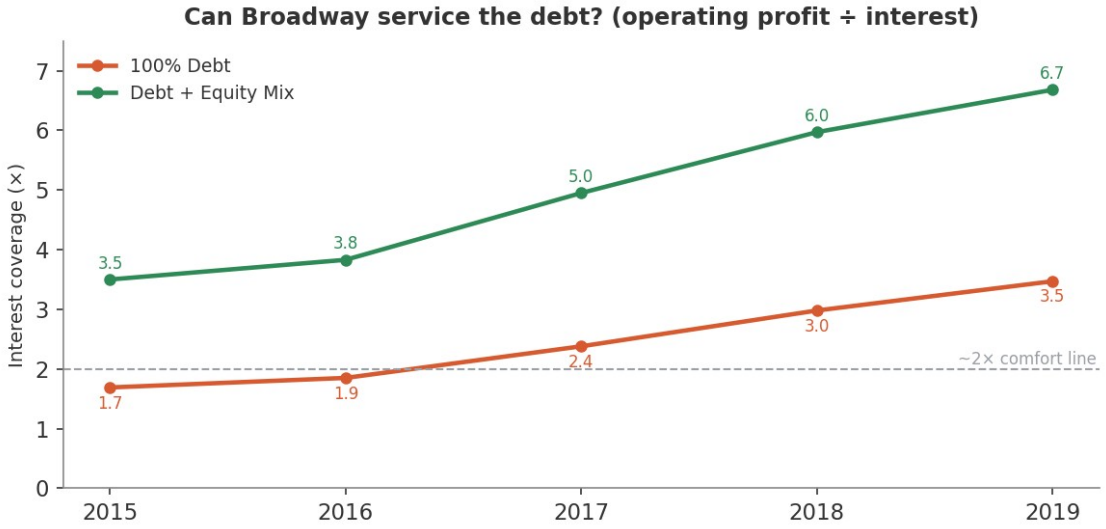
The debt-and-equity mix creates more than twice as much value for Broadway's existing shareholders.

Why the mix wins (this is the clever part): with 100% debt, existing owners keep all of the equity — but it's a small equity ($59.2m), because $127.9m of the firm's value is owed to lenders. With the mix, they keep only 60% — but of a much bigger equity ($119.2m), which comes to $71.5m. On top of that, the new investors pay $60m for a 40% stake that's really worth only $47.7m, effectively handing about $12m to the existing owners. So the mix leaves Harris and the current shareholders better off — and, as Module 6 shows, it's far safer.

A fair caveat: *this treatment holds the firm's value fixed and doesn't award the all-debt option a bigger tax shield. Give all-debt full credit for its extra interest tax shield and it looks a little better — but it also carries much more distress risk, which the coverage numbers below make real. On balance the mix is the sounder call.*

Module 6 — Can Broadway service the debt?

A deal that creates value on paper still fails if the company can't make its loan payments. The quickest test is the interest coverage ratio — operating profit ÷ interest. A ratio near or below 2 is uncomfortable; comfortably above 3 is safe.



Under 100% debt, coverage starts at just 1.7× in 2015 — below the comfort line, and dangerously thin if the turnaround stumbles. The mix starts at a comfortable 3.5×.

Two things stand out:

- **The all-debt plan is fragile early.** Year-1 coverage of 1.7× leaves almost no room for error — and this is exactly the period when the risky turnaround is unproven. It does improve as debt is repaid (3.5× by 2019), but the first two years are white-knuckle.
- **The mix is comfortable throughout.** Coverage of 3.5× rising to 6.7× — plenty of cushion.
- **Both plans need to refinance the balloon.** The $90m (all-debt, 2023) and $60m (mix, 2020) lump-sum repayments are far bigger than any single year's cash flow, so Broadway will have to roll them over — a normal but real risk.

The recommendation:

Do the deal — Landmark is worth ~$129.7m to Broadway, comfortably above the $120m price — but finance it with the debt-and-equity mix. It creates more value for existing shareholders ($22.0m vs $9.7m) and is far safer to service (3.5× coverage vs 1.7×), especially while the turnaround is still unproven.

Final results — all in one place

Item	Result
Discount rate (WACC)	8.68% (unlevered cost of capital 9.13%)
Landmark on its own	$26.0m
Improved Landmark (turnaround)	$100.9m
+ Improvements to Broadway	$28.8m
= Total value of Landmark to Broadway	**$129.7m**
Price	$120.0m
Value created — 100% Debt	$9.7m
Value created — Debt + Equity Mix	**$22.0m**
Year-1 interest coverage	1.7× (all-debt) vs 3.5× (mix)
Recommendation	**Buy Landmark; finance with the debt-and-equity mix**

Exam watch-outs

- **Unlever, average, relever.** Never use a comparable's raw beta — strip out its debt first (÷ [E+(1−t)D]), average, then re-load your firm's leverage.
- **Free cash flow = NOPAT + depreciation − change in working capital − capex.** NOPAT already has tax taken out; don't subtract interest (that's a financing item).
- **Value Landmark to Broadway, not on its own.** The whole point is that it's worth $26m alone but $129.7m to Broadway. The buyer-specific synergies justify the price.
- **Terminal value = next-year FCF ÷ (WACC − g).** Most of the value sits here; state your growth rate.
- **Financing splits a fixed pie.** Combined value is the same either way; more debt just means less equity. Compare value to EXISTING shareholders, not total equity.
- **The mix can beat all-debt.** Here the new investors overpay for their 40%, and the lighter debt is safer — so the mix creates more value for existing owners. Don't assume all-debt always wins.
- **Always check debt serviceability.** Interest coverage near/below 2× is a red flag — especially with an unproven turnaround. Note the balloon payments need refinancing.
- **Stress-test the assumptions.** The valuation lives or dies on the margin turnaround and the premium-pricing plan working. Say so.

End of study guide — good luck on Monday.